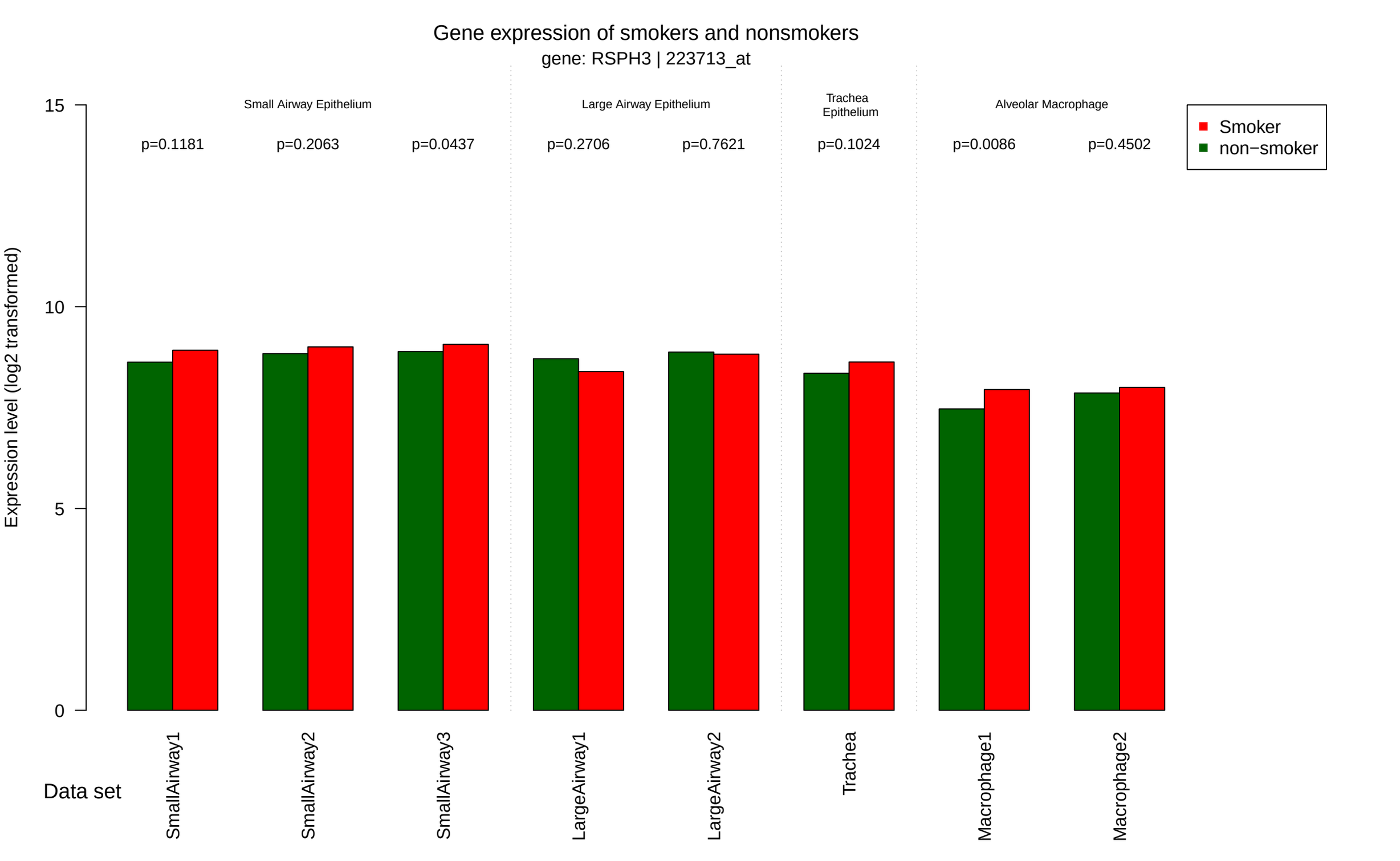
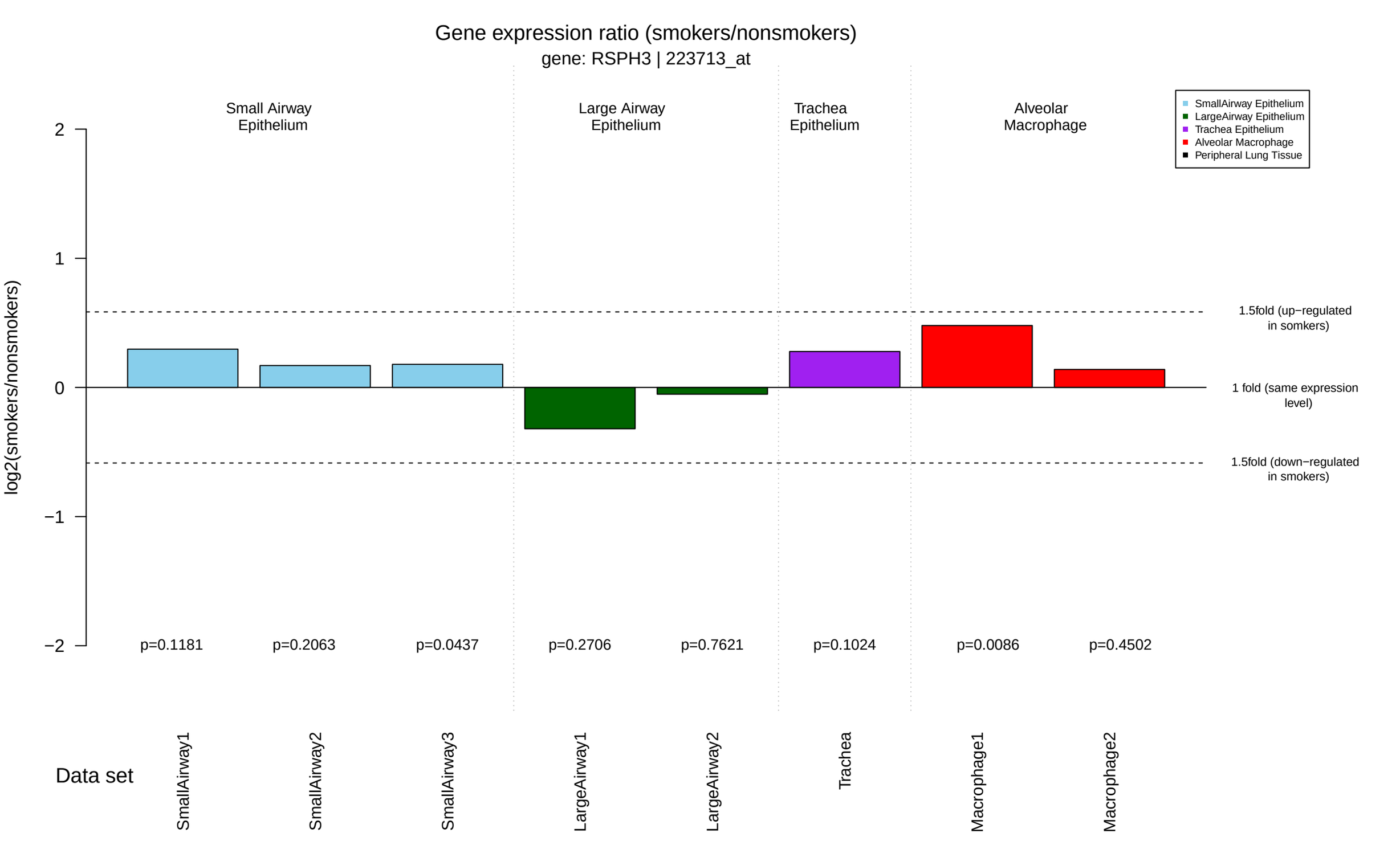
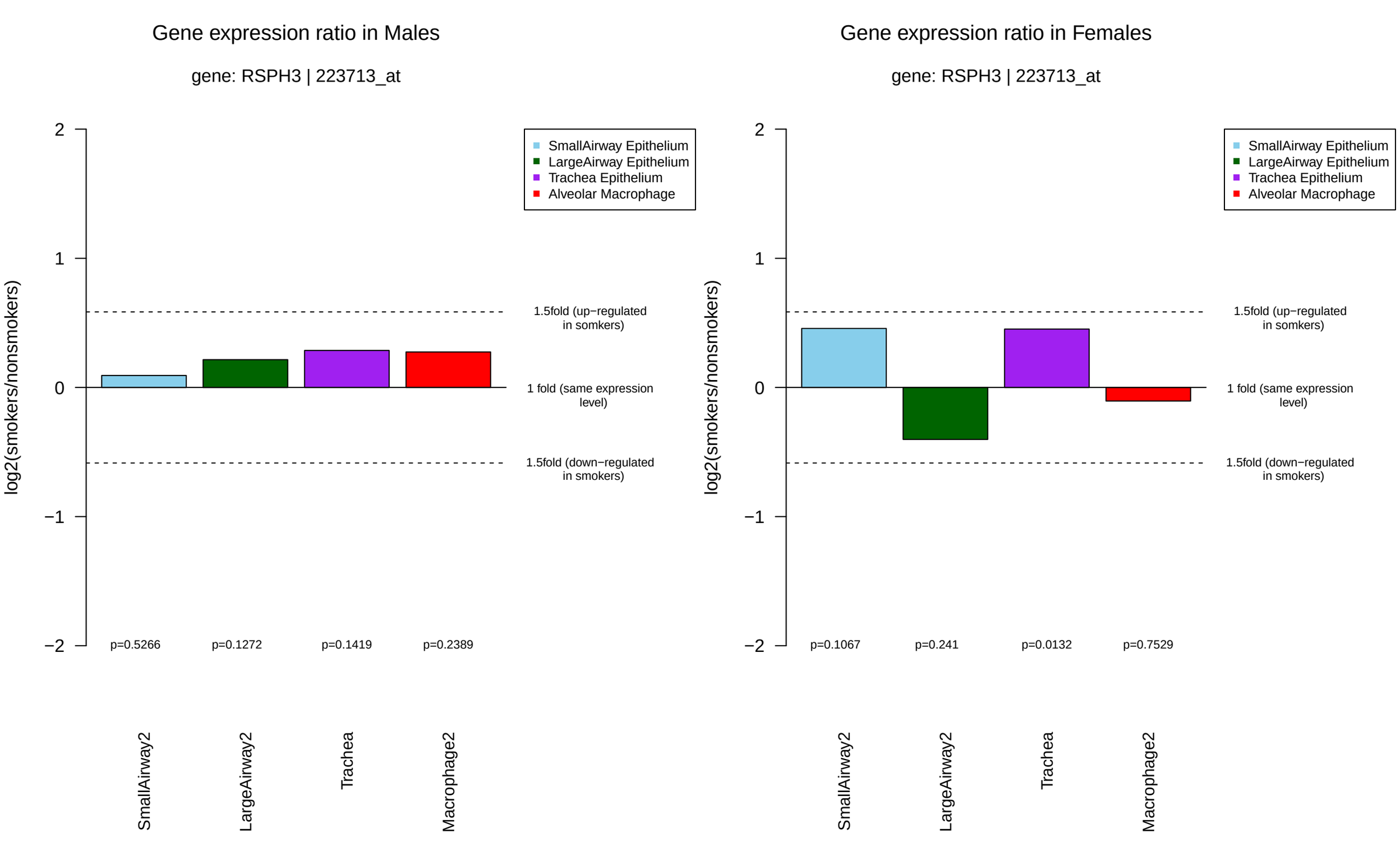
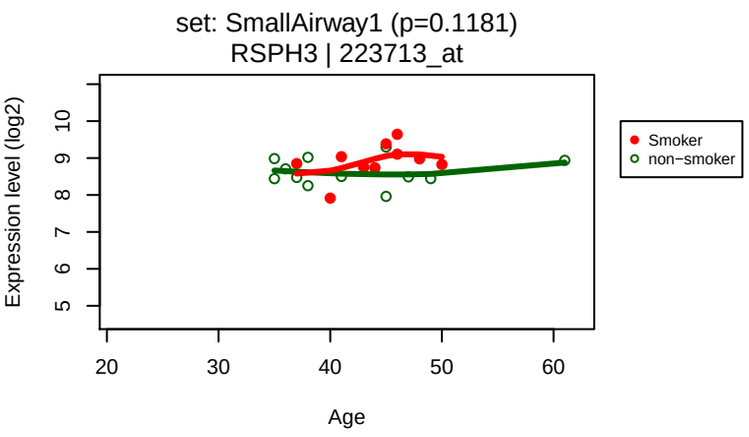
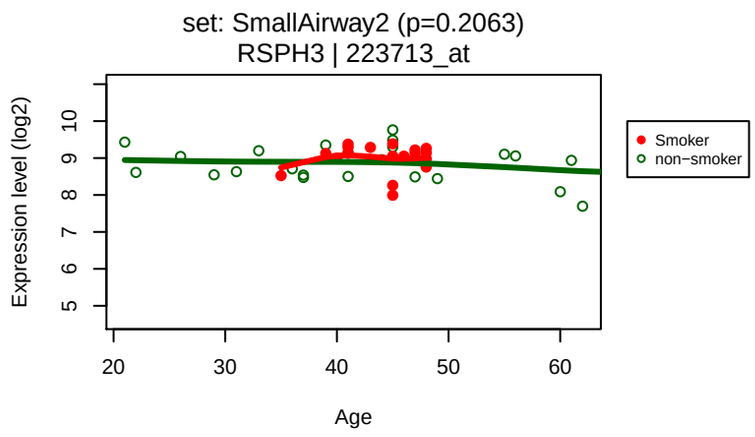
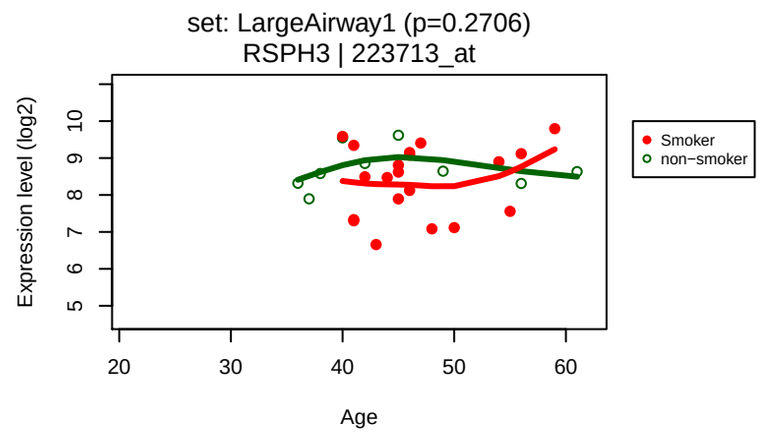

Gene expression of smokers and nonsmokers
gene: RSPH3 | 223713_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1181
p=0.2063
p=0.0437
p=0.2706
p=0.7621
p=0.1024
p=0.0086
p=0.4502
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

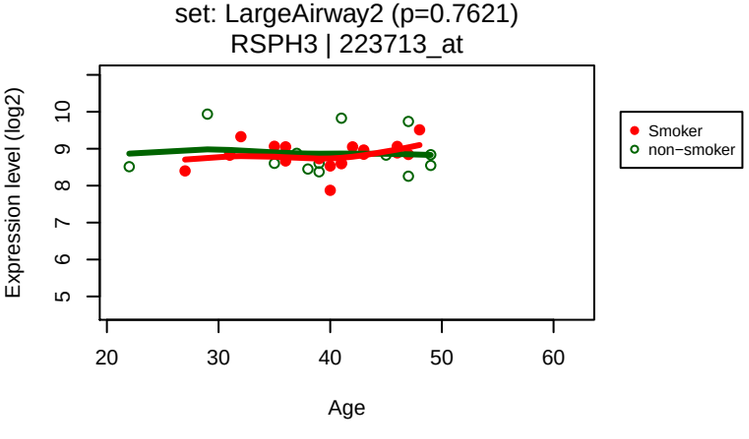

Gene expression ratio (smokers/nonsmokers)
gene: RSPH3 | 223713_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1181
p=0.2063
p=0.0437
p=0.2706
p=0.7621
p=0.1024
p=0.0086
p=0.4502
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

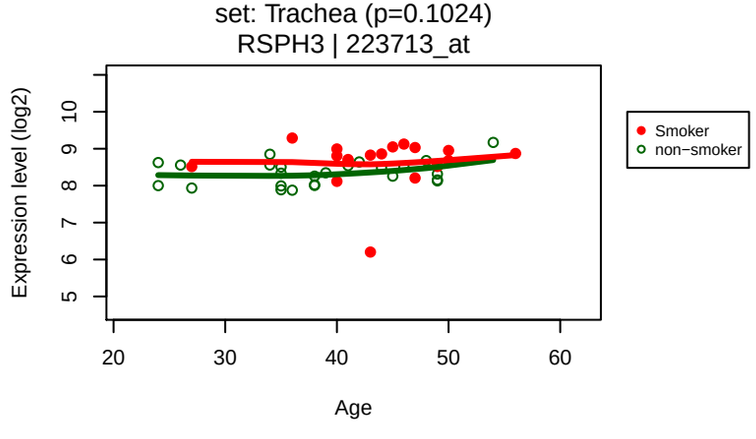

Gene expression ratio in Males
gene: RSPH3 | 223713_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5266
p=0.1272
p=0.1419
p=0.2389
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RSPH3 | 223713_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1067
p=0.241
p=0.0132
p=0.7529
SmallAirway2
LargeAirway2
Trachea
Macrophage2

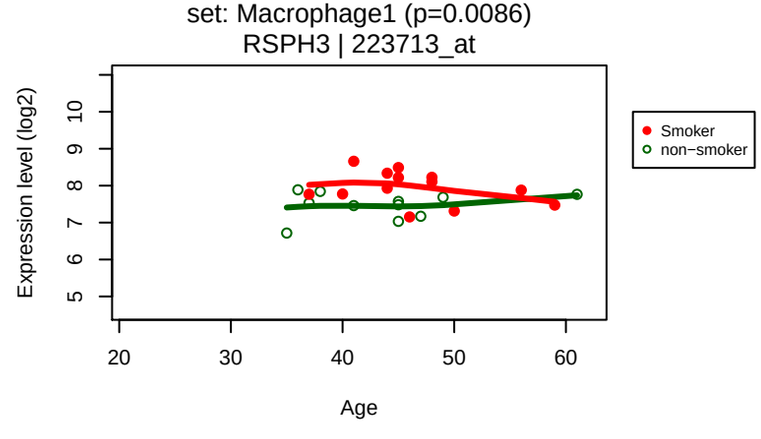

set: SmallAirway1 (p=0.1181)
RSPH3 | 223713_at
Expression level (log2)
Age
Smoker
non−smoker

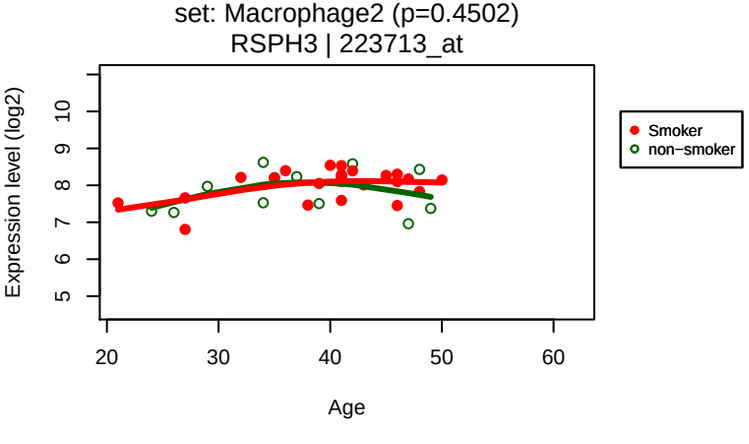

set: SmallAirway2 (p=0.2063)
RSPH3 | 223713_at
Expression level (log2)
Age
Smoker
non−smoker

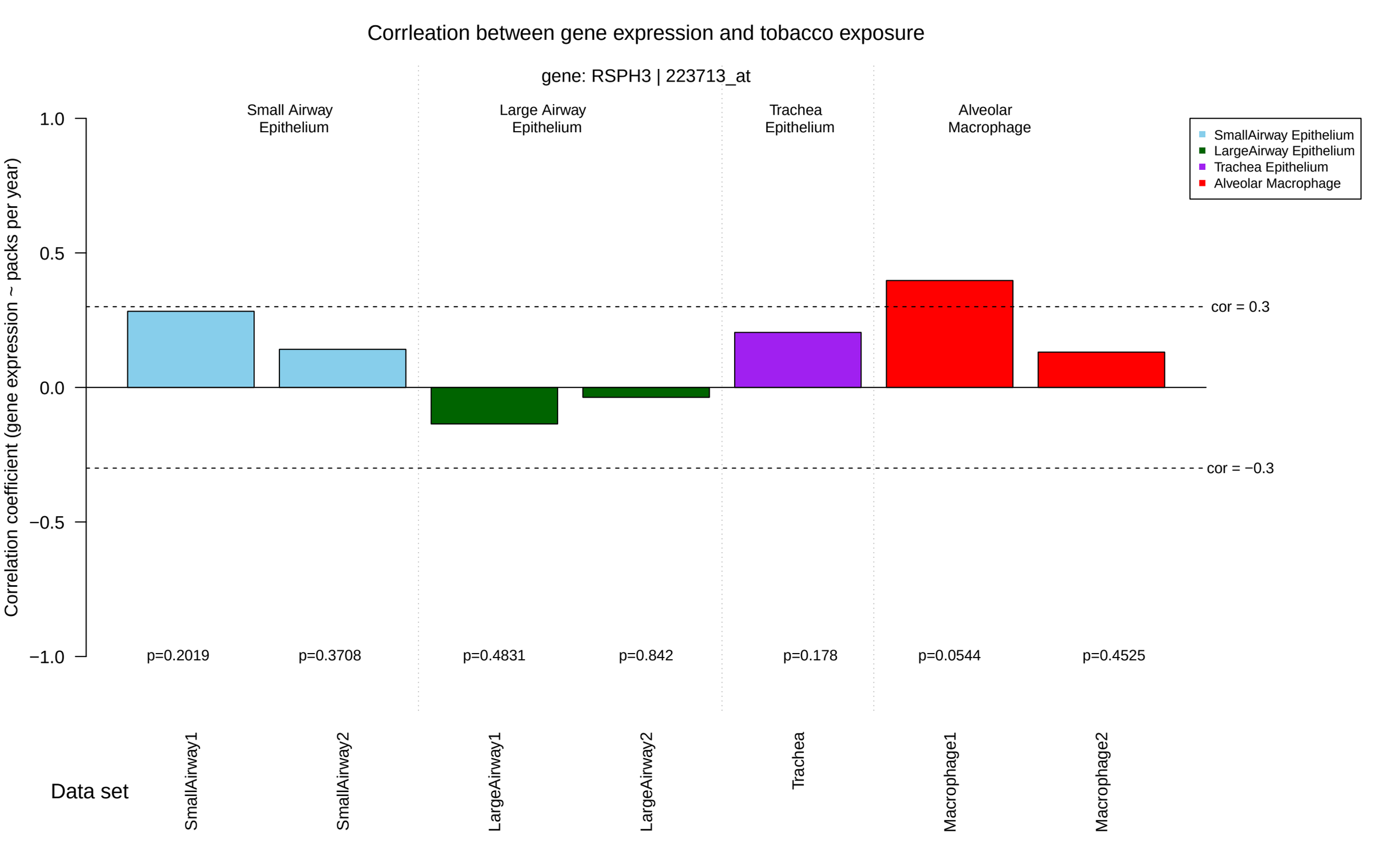

set: LargeAirway1 (p=0.2706)
RSPH3 | 223713_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7621)
RSPH3 | 223713_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1024)
RSPH3 | 223713_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0086)
RSPH3 | 223713_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4502)
RSPH3 | 223713_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RSPH3 | 223713_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2019
p=0.3708
p=0.4831
p=0.842
p=0.178
p=0.0544
p=0.4525
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2